

Secure ICS Telemetry & Detection Platform

Systems Security Team
Rolls-Royce Summer 2026



Project Goal

Develop a secure Industrial Control System (ICS) telemetry platform capable of:

- Collecting telemetry from simulated industrial devices
- Protecting communications using layered security controls
- Detecting cyber threats targeting operational technology environments
- Providing real-time security monitoring and visualization

Key Focus Areas

- Industrial Cybersecurity
- Zero Trust Architecture
- Telemetry Security
- Threat Detection
- Security Monitoring



Why This Matters

Industrial Control Systems are used in:

- Aerospace
- Manufacturing
- Energy
- Transportation
- Critical Infrastructure

Common Threats

- Spoofed telemetry
- Replay attacks
- Data tampering
- Unauthorized devices
- Denial of Service attacks

Project Objective

Create a secure telemetry environment capable of detecting and responding to these threats.



System Architecture

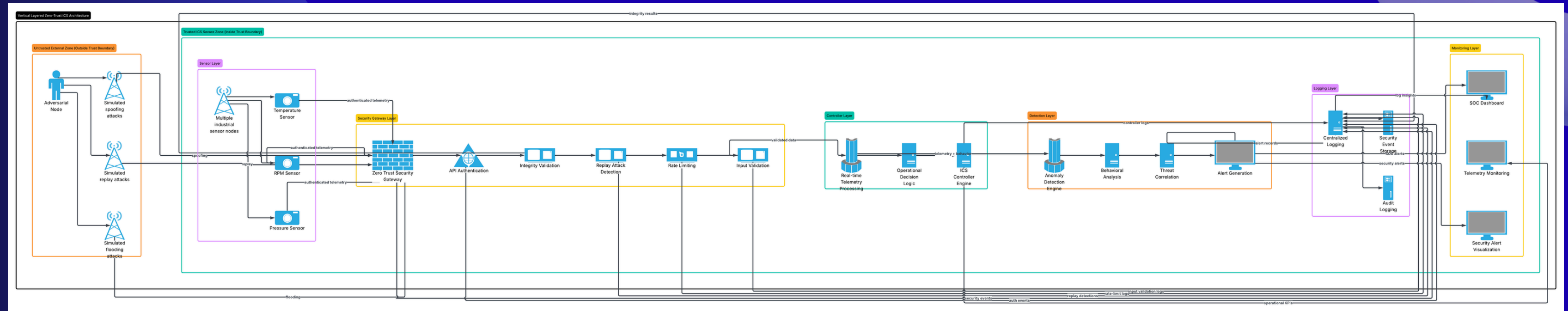


Figure 1: Secure ICS System Architecture

Week 1-2 Progress

Foundation & Threat Modeling

Completed:

- Project architecture design
- Threat model development
- Telemetry schema creation
- Zero Trust design principles
- Security requirements definition

Deliverables

- Architecture documentation
- Threat matrix
- Security requirements specification



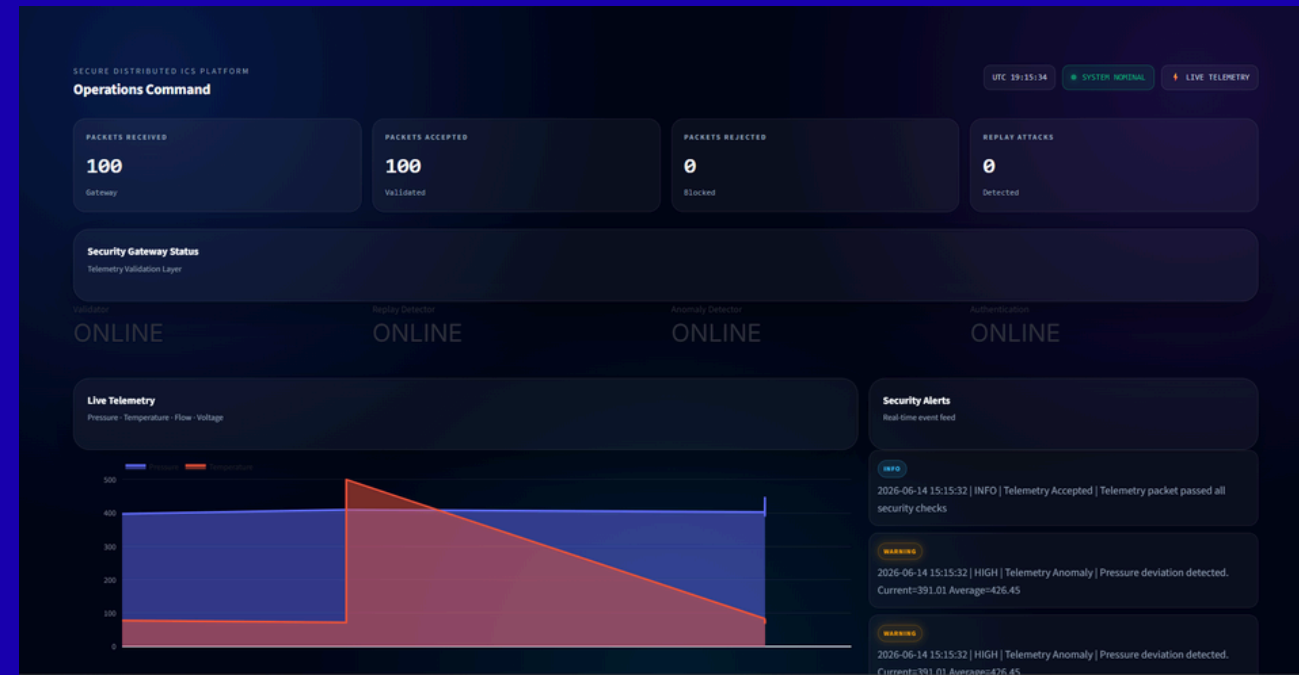


Week 3-4 Progress

Telemetry Pipeline & Dashboard

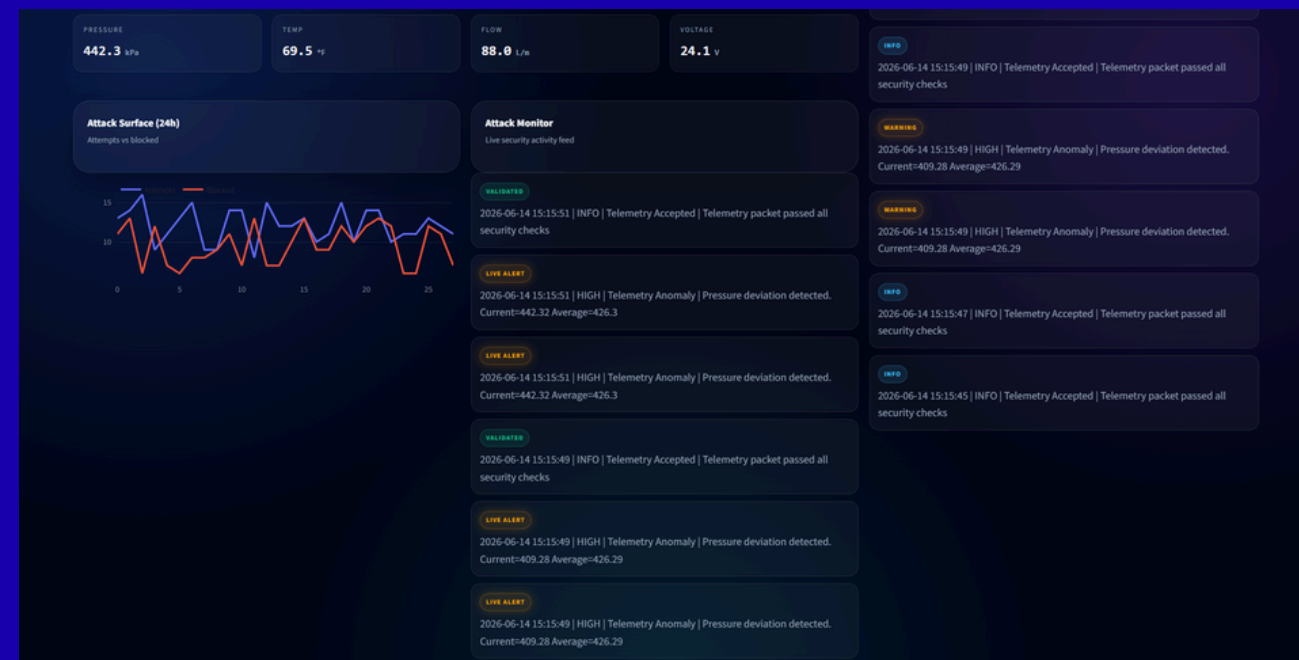
Completed

- Sensor simulation framework
- Telemetry generation
- Controller API design
- Dashboard development
- Real-time telemetry visualization



Dashboard Features

- Live telemetry monitoring
- Operational metrics
- Security event visibility



Operational Status				
ICS Device Fleet				
Device	Type	Zone	Load	Status
PLC-01	PLC	Zone A	38%	ONLINE
PLC-04	PLC	Zone A	81%	DEGRADED
RTU-12	RTU	Zone B	47%	ONLINE
Gateway-01	Gateway	DMZ	64%	ONLINE
Security Metrics				
Gateway Performance				
Anomalies	Replay Attacks	Rejected Packets	Accepted Packets	
6	0	0	100	



Week 5 Progress

Security Controls

Implemented:

Telemetry Validation

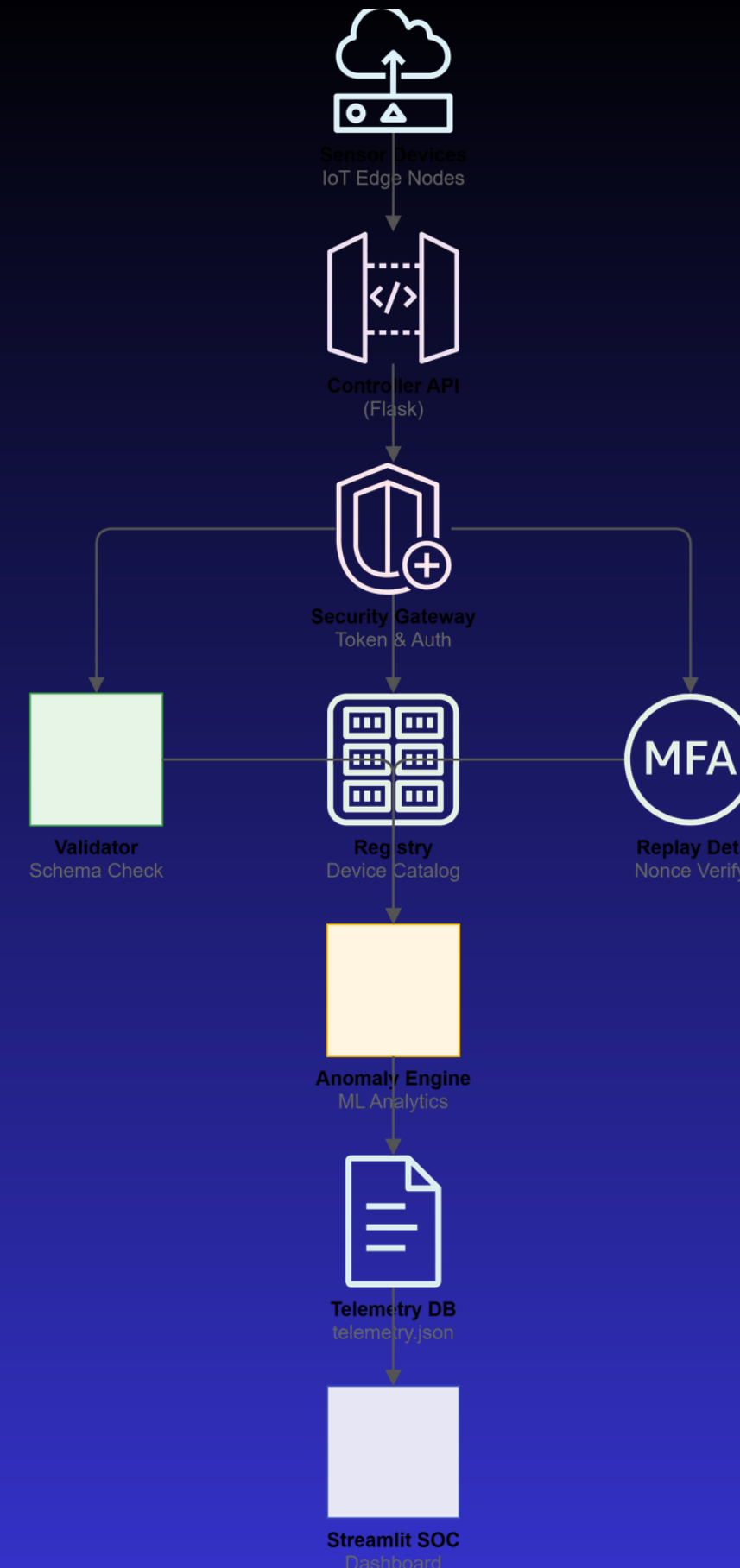
- HMAC SHA-256 packet signing
- Integrity verification

Replay Protection

- Timestamp validation
- Duplicate packet detection

Anomaly Detection

- Threshold monitoring
- Behavioral analysis





Week 6 Progress

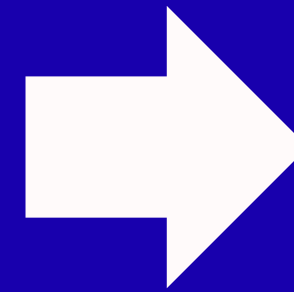
System Integration

Completed:

- Security Gateway integration
- Sensor Registry authentication
- Telemetry storage
- End-to-end telemetry pipeline
- Dashboard integration

Result:

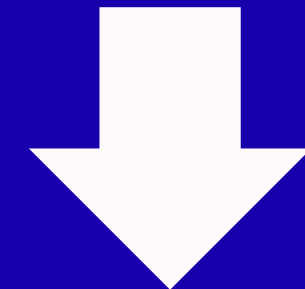
Sensor → API → Gateway → Storage → Dashboard
fully operational.



```
resp_iter = self.stub.GetStatusesStream()

statuses = {}
async for data in resp_iter:
    status = Status(
        status_id=data.id, name=data.name
    )
    statuses[status.name] = status

return statuses
```





Attack Simulations

Attack Framework

Implemented:

- Replay Attack
- Spoofing Attack
- Data Tampering Attack
- Flooding Attack
- Unauthorized Device Attack

Results

- Replay packets rejected
- Unauthorized devices blocked
- Tampered packets detected
- Security alerts generated

Attack Monitor

Live security activity feed

LIVE CRITICAL

2026-06-14 18:00:17 | HIGH | Replay Attack | Replay attack detected

LIVE CRITICAL

2026-06-14 18:00:17 | HIGH | Replay Attack | Replay attack detected

LIVE CRITICAL

2026-06-14 18:00:17 | HIGH | Replay Attack | Replay attack detected

VALIDATED

2026-06-14 18:00:17 | INFO | Telemetry Accepted | Telemetry packet passed all security checks

LIVE ALERT

2026-06-14 18:00:04 | HIGH | Unauthorized Sensor | Unauthorized device attempted communication: hacker_device

LIVE CRITICAL

2026-06-14 17:59:39 | CRITICAL | Integrity Failure | Telemetry signature mismatch



Current Platform Capabilities

Operational Features

- ✓ Live Telemetry Monitoring
- ✓ Security Event Monitoring
- ✓ Device Authentication
- ✓ Integrity Validation
- ✓ Replay Protection
- ✓ Behavioral Anomaly Detection
- ✓ Attack Simulation Framework
- ✓ Centralized Logging

Challenges & Lessons Learned

Challenges

- Integrating multiple security modules
- Python package management
- Live telemetry pipeline development
- Dashboard synchronization

Lessons Learned

- Importance of layered security controls
- Secure telemetry architecture design
- OT cybersecurity concepts
- Real-time monitoring system development





Next Steps (Weeks 7-12)

Planned Enhancements

- Security analytics
- Historical trend analysis
- Threat scoring
- Automated reporting
- Incident response workflows
- Dashboard improvements
- Advanced detection capabilities

Goal

Transform the platform into a SOC-style OT monitoring environment.





Questions

Thank you

Any Questions?